Tatsuya Terao

Doctoral Student

Research Institute for Mathematical Sciences, Kyoto University, Kyoto 606-8502, Japan

ttatsuya@kurims.kyoto-u.ac.jp | otera99.github.io/

Research Interests

- Theoretical Computer Science and Algorithm Design.

Education

Kyoto University *Kyoto, Japan*

Bachelor of Science *April 1, 2018 - March 31, 2022*

- Faculty of Science, Division of Physics

Kyoto University *Kyoto, Japan*

Master of Science *April 1, 2022 - March 31, 2024*

- Advisor: Yusuke Kobayashi

Kyoto University *Kyoto, Japan*

Doctor of Science *April 1, 2024 - present*

- Advisor: Yusuke Kobayashi

Publications

As is the convention in theoretical computer science, authors are listed alphabetically, except that Ⓒ and Ⓡ are used in place of commas to indicate contribution order and random order, respectively.

1. Yusuke Kobayashi and Tatsuya Terao: One-Face Shortest Disjoint Paths with a Deviation Terminal,
   In Proceedings of the 33rd International Symposium on Algorithms and Computation (**ISAAC 2022**), 47:1–47:15.
   doi:10.4230/LIPIcs.ISAAC.2022.47

2. Tatsuya Terao: Faster Matroid Partition Algorithms,
   A preliminary version appeared in Proceedings of the 50th EATCS International Colloquium on Automata, Languages and Programming (**ICALP 2023**), 104:1–104:20.
   doi:10.4230/LIPIcs.ICALP.2023.104
   ACM Transactions on Algorithms (**TALG**), Volume 21, Issue 2, 2025.
   doi:10.1145/3707208

3. Yusuke Kobayashi and Tatsuya Terao: Subquadratic Submodular Maximization with a General Matroid Constraint,
   In Proceedings of the 51st EATCS International Colloquium on Automata, Languages and Programming (**ICALP 2024**), 100:1–100:19.
   doi:10.4230/LIPIcs.ICALP.2024.100

4. Tatsuya Terao Ⓒ Ryuhei Mori: Parameterized Quantum Query Algorithms for Graph Problems,
   In Proceedings of the 32nd Annual European Symposium on Algorithms (**ESA 2024**), 99:1-99:16.
   doi:10.4230/LIPIcs.ESA.2024.99

5. Tatsuya Terao: Deterministic $(2/3-\varepsilon)$-Approximation of Matroid Intersection Using Nearly-Linear Independence-Oracle Queries,
   In Proceedings of the 19th Algorithms and Data Structures Symposium (**WADS 2025**), 50:1–50:18.
   doi:10.4230/LIPIcs.WADS.2025.50

6. Chien-Chung Huang, Naonori Kakimura, Yusuke Kobayashi, Tatsuya Terao: Polynomial Kernels with Reachability for Weighted d-Matroid Intersection,
   In Proceedings of the 27th Conference on Integer Programming and Combinatorial Optimization (**IPCO 2026**), 362–377.
   doi:10.1007/978-3-032-28691-8_24

7. Tatsuya Terao: Faster Approximate Linear Matroid Intersection,
   In Proceedings of the 20th Scandinavian Symposium on Algorithm Theory (**SWAT 2026**), 39:1–39:19.
   doi:10.4230/LIPIcs.SWAT.2026.39
   Best Student Paper Award.

8. Yonggang Jiang and Tatsuya Terao: Almost Optimal Streaming and Cut-Query Algorithms for Undirected Max-Flow, To appear in Proceedings of the 38th ACM-SIAM Symposium on Discrete Algorithms (**SODA 2027**).

Awards

June 19, 2026 **Best Student Paper Award**, 20th Scandinavian Symposium on Algorithm Theory (**SWAT 2026**)

Grants

2024-2027 Research Fellowships for Young Scientists (DC1), Japan Society for the Promotion of Science

Presentations

- One-Face Shortest Disjoint Paths with a Deviation Terminal.
  - ISAAC 2022, Seoul, Korea, Dec. 20, 2022.
- Faster Matroid Partition Algorithms.
  - ICALP 2023, Paderborn, Germany, July 14, 2023.
  - 13th Hungarian-Japanese Symposium on Discrete Mathematics and Its Applications, Tokyo, Japan, May 26, 2025.
- Subquadratic Submodular Maximization with a General Matroid Constraint.
  - ICALP 2024, Tallinn, Estonia, July 9, 2024.
- Parameterized Quantum Query Algorithms for Graph Problems.
  - ESA 2024, Egham, United Kingdom, Sep. 4, 2024.
- Deterministic $(2/3 - \varepsilon)$-Approximation of Matroid Intersection Using Nearly-Linear Independence-Oracle Queries
  - WADS 2025, Toronto, Canada, Aug. 13, 2025.

Services

Peer Review

I've been a subreviewer for the conference: RANDOM 2025, IPCO 2026, ICALP 2026, MFCS 2026, SODA 2027, ITCS 2027, STACS 2027.